

Name _____ Index No. _____ / _____

2920/302
MANAGEMENT INFORMATION SYSTEMS
November 2015
Time: 3 hours

Candidate's Signature _____

Date _____



THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY
MODULE III

MANAGEMENT INFORMATION SYSTEMS

3 hours

INSTRUCTIONS TO CANDIDATES:

Write your name and index number in the spaces provided above.

Sign and write the date of examination in the spaces provided above.

*Answer any **FIVE** of the following **EIGHT** questions on the spaces provide on the question paper.*

All questions carry equal marks.

Candidates should answer the questions in English.

For Examiners' Use only

Questions	1	2	3	4	5	6	7	8	Total Score
Candidate Score									

This paper consists of 16 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

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Turn over

1. (a) Outline **four** features of an executive information system (EIS). (4 marks)

(b) Define each of the following terms as applied in management information systems:

- (i) interaction; (2 marks)

[illegible]

- (ii) interdependence. (2 marks)

- (c) The management of KPNG Company has purchased a transaction processing system for its operations. Explain **three** functions of the system to the company. (6 marks)

[illegible][illegible]

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- (d) Explain **three** circumstances under which an organization would adopt the use of expert systems in its operations. (6 marks)

2. (a) Explain each of the following terms as used in project management:

- (i) project planning document (PPD); (2 marks)

- (ii) total quality management (TQM). (2 marks)

- (b) The system analyst of ABX Company has recommended for modification on their current MIS. Explain **three** factors that could have contributed to this move. (6 marks)

(c) Differentiate between a *Trojan horse* and a *worm* as used in computer security. (4 marks)

(d) With the aid of a diagram, describe the components of a decision support system (DSS). (6 marks)

3. (a) Explain each of the following modes of data processing as applied in management information systems:

(i) real-time transaction;

(2 marks)

(ii) batch.

(2 marks)

(b) Differentiate between *efficiency* and *effectiveness* as used in management information systems.

(4 marks)

(c) Project budgets could be implemented using the top-down approach in an organization.

(i) Explain **two** advantages of this approach.

(4 marks)

(ii) Outline **two** disadvantages of the approach.

(2 marks)

(d) Explain **three** strategies that project managers could use to manage risks.

(6 marks)

4. (a) Explain each of the following terms as used in organizational change:

(i) rationalization;

(2 marks)

(ii) business process re-engineering.

(2 marks)

- (b) Differentiate between a *hacker* and a *cracker* as used in computer security. (4 marks)

- (c) Jabil Company has incorporated an information system for its decision making process. Explain **three** ways that the system would help improve the decision making process. (6 marks)

[illegible]

- (d) Explain **three** circumstances that would cause an organization to in cooperate an information system in the production section. (6 marks)

[illegible]

- (4 marks)

(b) Jade intends to write a term paper on ways in which system users could get involved in unethical use of information. Explain **three** such ways. (6 marks)

- (c) With the aid of a diagram describe the decision making model as used in management information system. (4 marks)

- (d) Reki Company intends to acquire an effective human resource management system for its operations. Explain **three** factors that they should consider. (6 marks)

6. (a) Explain the following terms as used in management information systems:

- (i) computer aided design; (2 marks)

- (ii) computer aided manufacture. (2 marks)

- (b) Companies could acquire their information system through leasing method.

- (i) Outline **three** advantages of this method. (3 marks)

(ii) Outline **three** limitations of the method.

(3 marks)

(c) Explain **three** challenges experienced in regulating global information society. (6 marks)

(d) The analyst of Zippra Company has advised the company to acquire a security management system for its operations. Explain **two** functions of this software to the company. (4 marks)

7. (a) Define each of the following types of maintenance as used in management information systems:

(i) corrective;

(2 marks)

(ii) preventive.

(2 marks)

(b) Planning is considered as an important function of an MIS in management. Explain **three** importance of this function to an organization. (6 marks)

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- The diagram illustrates a multi-layer PCB layout, showing the top, middle, and bottom layers. The top layer contains various components, including a microcontroller (U1), a memory chip (U2), a power management IC (U3), and several passive components like resistors and capacitors. The middle layer shows a complex network of traces connecting these components. The bottom layer features a large rectangular area labeled 'BOTTOM LAYER' and a smaller area labeled 'BOTTOM LAYER'. The layout is organized into functional blocks, with a central processing area and peripheral control areas.

- [illegible]

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(iii) outline two advantages of the above method in project management.

(2 marks)